

Cold-Start Ranking: Architectural Semantic Labels Predict Memory Use Where Graph Methods Go Blind

Abstract

In agent memory systems, graph-based ranking methods go blind at cold-start: 91% of new nodes remain invisible to PageRank and access-count scores throughout the observation window. Architectural semantic labels assigned at node creation predict future organic use at AUC 0.84, a 34-point advantage over random, of which approximately 18 points exceed pure-text baselines (0.66). We document this in a self-built agent research testbed, in operation since March 2026. Cold-start is persistent: after an August 2026 architectural transition, 91% of new nodes had not accumulated a single access seven weeks later (October 2026). The newest-first baseline performs below random (AUC 0.41), because use accrues with time. The advantage is architectural: creation-time categorical assignments (semantic domain, content category, provenance type) carry signal that shallow lexical features cannot extract. With accumulated history, structural reinforcement takes over (AUC 0.917); whether labels add increment there was not tested. We contribute a quantified cold-start blind spot, architectural labels as cold-start priors, a label-versus-text disentanglement, and a stated boundary: the advantage is regime-dependent and partially circular. No cross-system superiority is claimed.

1 Introduction

Agent memory systems face a problem that recommendation systems have long studied but agent memory architectures have not yet quantified: **what do you do with something new?**

When a memory node is created in a long-running agent system—a conversation fragment, a derived principle, a self-narration—the node enters the store with zero history. No one has accessed it. No edges connect it to the graph. Every ranking signal that depends on accumulated use—access counts, graph centrality, collaborative filtering—returns a zero or a constant (Figure 1). In our testbed, 91% of nodes created after an architectural transition remain cold-start seven weeks after the transition.

The industry answer, borrowed from recommendation systems, is graph-based ranking (PageRank, degree centrality, link analysis). These methods work when the graph has edges. For a

node with zero in-degree, PageRank returns a constant—the method is mathematically undefined in this domain.

We document a different answer that grew in the system rather than being bolted on: **architectural semantic labels assigned at creation time**. In our testbed, every node receives categorical assignments when it enters the store—a semantic domain (chat, core, project, knowledge), a content category (self-narration, plan, rule, finding), a provenance type (raw user input, derived, identity). 74% of these labels are assigned by a derivation pipeline (inheriting categories from parent nodes); the remainder are assigned by architectural rules at creation. They are available at creation time—before any use; but their informational content derives from the system's categorization scheme, not solely from the node's raw text.

The key finding (Figure 2): **these labels predict future organic use at AUC 0.84, while PageRank sits at 0.50 (random) and newest-first at 0.41 (below random)**. The signal is real, it is available at creation time (before any use), and it is not extractable from the node's text content alone.

This paper is the third in a series. The first examined what to remember (signal separation in memory systems); the second examined when to stop (termination control as a database property). This one examines how to rank what is new. The three share a testbed but address orthogonal decisions.

2 Definitions

Cold-start node. A memory node that has been created but has accumulated zero access count through the observation window. In our testbed, 91.1% of nodes created after a 2026-08-15 architectural transition remain cold-start (originally measured at 92% in September; the October live read confirms at 91.1%). This is not a transient phase; it is the steady state for most nodes.

Architectural semantic label. A categorical assignment made at node creation (74% inherited via the derivation pipeline, the remainder by creation-time rules), reflecting the node's semantic role: semantic domain (where does this belong), content category (what kind of content is this), provenance type (where did this come from). These are architectural decisions, not learned features.

Organic use. A node being referenced, linked, or retrieved by the system's normal operation, measured after a grace period (minimum 3 days) to exclude construction-time edges.

Operating regime. A period during which the system's configuration, labeling taxonomy, and usage patterns are relatively stable. Regime changes occur when the architecture evolves significantly.

3 Related Work

Cold-start ranking is well-studied in recommendation systems, where content-based filtering assigns scores to new items using their creation-time attributes; the direct precedent for our approach. Recent work uses LLMs to simulate interactions for cold items (ColdLLM). Our contribution is not the mechanism (labels-as-priors is standard) but the **quantification** in an agent-memory context: a measured blind spot, a controlled prediction task, and a label-text disentanglement.

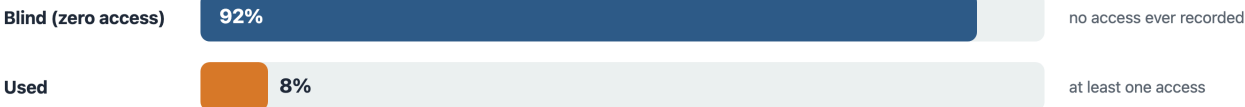
In agent memory, A-MEM generates structured attributes (descriptions, keywords, tags) at note creation: creation-time semantic assignment already exists. SYNAPSE uses spreading activation for memory ranking. The Generative Agents memory system combines relevance, recency, and importance; the last being an LLM-assigned score. None of these quantify cold-start prediction power, compare against graph baselines, or disentangle label from text contribution. That measurement gap is what this paper fills.

4 The Cold-Start Blind Spot

The cold-start blind spot

New nodes in the testbed enter the store with zero history — and mostly never accumulate any.

SEPTEMBER 2026 SNAPSHOT — 15,751 NODES CREATED AFTER THE ARCHITECTURAL TRANSITION



October 2026 live read confirms it persists: **91.1% blind** across 19,568 post-transition nodes.

GRAPH-BASED AND REGENCY RANKING ON THESE COLD-START NODES

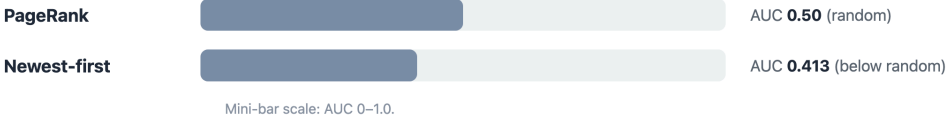


Figure 1: Cold-start is the norm: 91.1% of new nodes remain invisible to graph-based ranking throughout the observation window.

The blind spot is measurable.

After the 2026-08-15 transition, 15,751 new nodes were created. By the measurement date, 1,289 had at least one access (access count > 0). **The remaining 14,462 (92%) had zero.** A further measurement on the live store (October 2026) confirms: 19,754 nodes created post-transition, 1,747 with use (91.16% blind).

For these cold-start nodes, every use-dependent signal is zero or constant:

Table 1: Ranking signals and their cold-start behavior (post-transition cohort, September snapshot).

Signal	Value for cold-start nodes	AUC
Access count	0 for 92%	undefined
PageRank	constant (zero in-degree)	0.50 (mathematical fact)
Degree centrality	0 or construction-only	≈ random
Newest-first (recency)	higher for newer nodes	0.41 (below random)

The recency result deserves note: sorting by newest-first performs **below** random because use accrues with time. A node created yesterday has had less opportunity for use than one created last month. The trivial heuristic is anti-predictive.

5 Architectural Labels as Cold-Start Priors

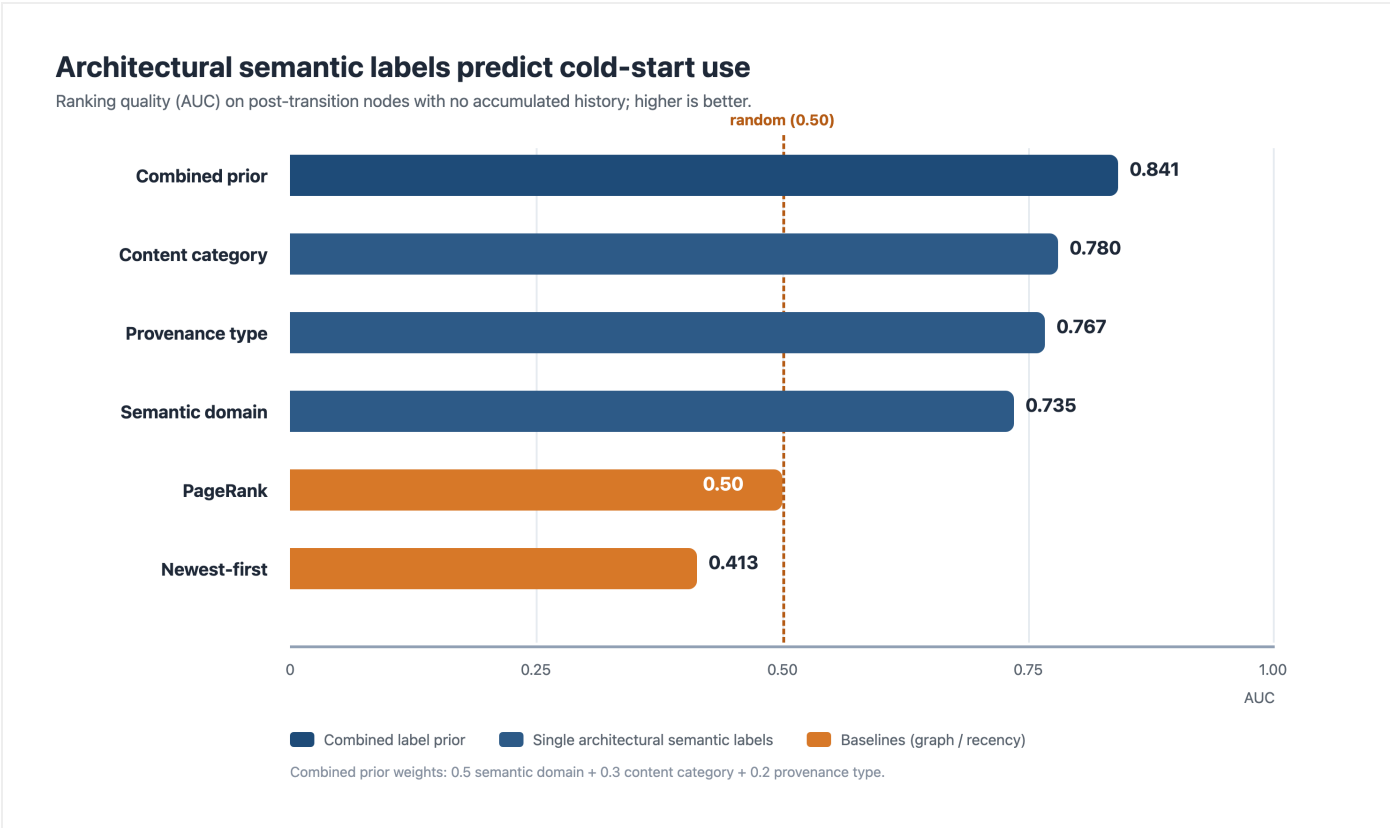


Figure 2: Architectural semantic labels predict cold-start use at AUC 0.841; graph and recency baselines are blind.

If use-dependent signals are blind, what signal exists at creation time?

The architecture's own labeling system. Every node receives three categorical labels at creation. These labels partition the node space into semantically meaningful groups; and some groups are systematically more likely to be used than others.

We tested this with a grace-period-controlled cold-start prediction task: for nodes created in a training window, learn the prior probability of organic use for each label value; predict for nodes in a subsequent test window.

Results (14-day organic use, training 2026-07-01 to 08-15, testing 08-15 to 09-10):

Table 2: AUC by prior type with 95% CI (A-window evaluation).

Prior type	AUC	95% CI
Semantic domain	0.735	[0.715, 0.752]
Content category	0.780	[0.762, 0.794]
Provenance type	0.767	—

Combined (0.5 dom + 0.3 cat + 0.2 prov)	0.841	[0.821, 0.861] ¹
PageRank	0.50	(random)
Newest-first	0.413	(below random)

The combined prior's P@100 is 0.84: among the top 100 ranked cold-start nodes, 84% are subsequently used.

Grace-period sensitivity. The signal strengthens as construction-time edges are excluded: 5-minute grace 0.735, 1-hour 0.766, 1-day 0.826, 3-day 0.849. This confirms the prediction captures organic use, not construction artifacts.

6 What the Advantage Is Not

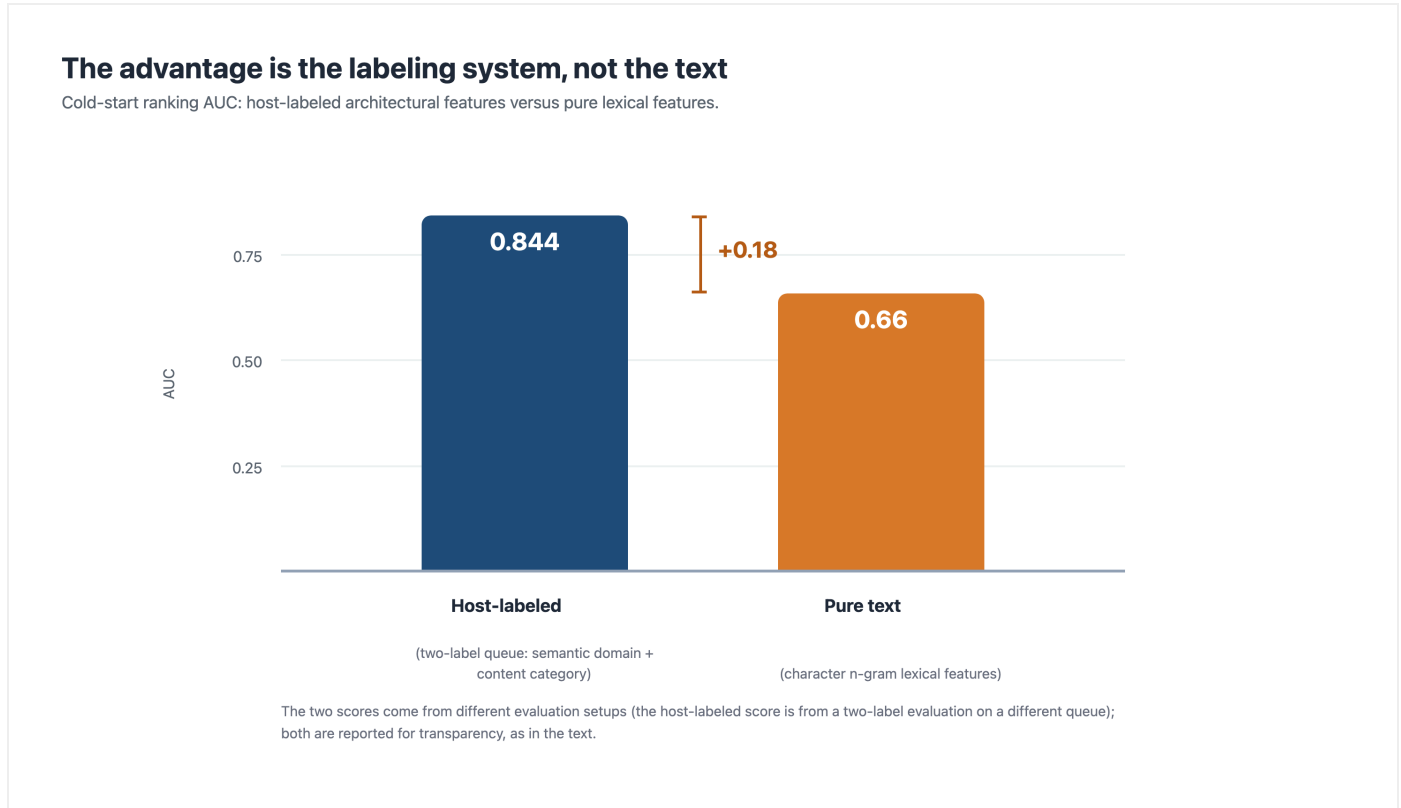


Figure 3: The advantage is in the labeling system, not the text: a +0.18 gap beyond pure lexical features.

Four candidate explanations are examined and excluded in turn (Sections 6.1–6.4): text content, recency, graph structure, and partial circularity.

6.1 Not text content

Pure lexical features (character 2-3gram Hashing + k-means/LogReg) reach only AUC 0.657–0.665. Host-labeled features (semantic domain + content category, from a two-label evaluation on a different queue) reach 0.844. ¹The combined prior and host-labeled scores come from different evaluation setups; we report both for transparency. The gap of 0.18 is the labeling system's contribution, not the text's (Figure 3).

A reviewer might ask whether modern embeddings or LLM zero-shot scoring would close the gap. Our character n-gram baseline is admittedly shallow; stronger text baselines (embeddings, LLM zero-shot) may narrow the gap. The claim is therefore not that text is uninformative, but that the system's **categorization scheme**, a structured taxonomy built over seven months of operation, adds signal beyond what generic text features capture. The advantage is in the taxonomy, not in any individual label.

6.2 Not recency

Newest-first: AUC 0.413 (below random). Use accrues with time, not against it.

6.3 Not graph structure

PageRank: AUC 0.50 for zero-in-degree nodes (mathematical fact). Construction-time edges could be allowed for graph methods, but this gives graph methods a signal the labels don't use; making the comparison unfair in the graph's favor, not ours.

6.4 Partially circular

We disclose that semantic domain has a small weight in the system's ranking function (same-domain bonus 1.05×). This creates partial circularity: the system slightly favors certain domains in retrieval, which inflates the apparent predictive power. The content-category boost was explicitly removed by the architect as over-engineering. We cannot precisely bound the circularity contribution; the below-random recency result suggests use is not driven by recency preference, but a label-blind verification (e.g., using ephemeral edges as an independent use signal) remains future work.

7 With History, Labels: Untested

For nodes with accumulated history, structural reinforcement (in-degree from reinforce-type edges) achieves AUC 0.917. Whether the label prior adds any increment in this regime was not

tested (no within-regime comparison was run); the claim of this paper is limited to the cold-start regime.

This is the boundary of the paper's claim. Architectural labels are a **cold-start solution**, not a general ranking improvement. With history, structure takes over.

8 Rolling Priors

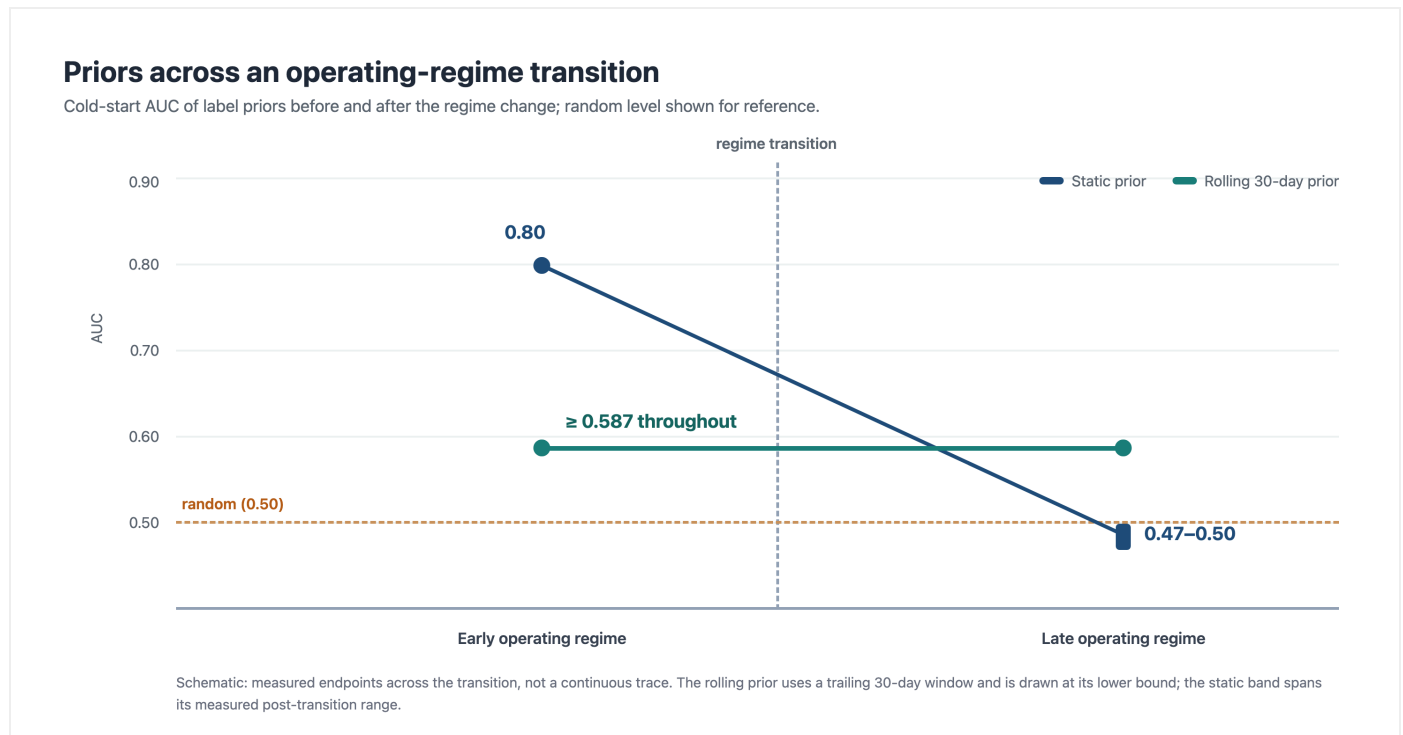


Figure 4: Static priors crash at regime transitions; rolling 30-day priors maintain signal.

and Regime Dependence

Static priors crash when the operating regime changes (Figure 4): AUC drops from 0.80 to 0.47–0.50 across a regime transition. Rolling priors (trailing 30-day window) maintain $\text{AUC} \geq 0.587$ throughout.

The regime dependence is directional: training on the early regime and testing on the late regime works (0.80); training on the late regime and testing on the early regime does not (0.50, random). This asymmetry means the label taxonomy itself evolves; later categories are richer and more predictive within their own regime but don't transfer backward.

For practitioners: cold-start priors must be **rolling**, not static. The mapping from labels to use-probability drifts as the system evolves.

9 Limitations

Single system, single user, closed-source testbed. Observational, not causal (a three-arm intervention was planned but not executed). 74% of cold-start labels come from the derivation pipeline rather than direct architectural assignment. Semantic domain has a small same-domain weight in ranking (partial circularity). The text baseline is shallow (character n-grams, not embeddings or LLM scoring). Regime-dependent (labels don't transfer backward across regime changes). No external replication. The cold-start prediction results are from a single evaluation window (A-source); independent B-window recomputation is pending. No external recall-assist mechanisms were active during the observation window; such mechanisms exist in the architecture but were disabled throughout, making this a clean baseline measurement.

10 Conclusion

Cold-start is the norm in agent memory: 91% of new nodes never accumulate a single access. Graph methods are mathematically blind in this domain. But the architecture's own labeling system, the categorical assignments it makes when a node enters the store, predicts future use at AUC 0.84, a signal that is available at creation time, not extractable from text, and not replicable by trivial heuristics. With history, structure takes over; the label increment there remains untested. The architecture's judgment about what a node is turns out to be the best predictor of what the node will *do*.

Appendix A: Series Division

This paper is third in a series sharing a testbed but addressing orthogonal decisions:

Table A1: Division of labor across the three-paper series.

	Paper 1 (#12919)	Paper 2 (#12935)	This paper
Decision	what to remember	when to stop	how to rank what is new
Signal timing	at query	at termination	at creation
Core taxonomy	salience/ranking/metabolism	model/framework/data-layer	graph/labels/recency
Key metric	semantic skew 93×	convergence 29 rounds	AUC 0.84 vs 0.50

Appendix B: Metric Definitions

- **AUC:** Area under the ROC curve for binary classification (used vs. not-used). Chance = 0.50.
- **P@100:** Precision at rank 100. Among the top 100 ranked nodes, the fraction subsequently used.
- **Grace period:** Time after node creation during which edges are excluded as construction artifacts. Main results use 3-day grace unless noted.
- **Organic use:** access count > 0 after grace period.
- **Rolling prior:** Prior probability estimated from the trailing 30-day window, with Laplace smoothing.

Version: v1.1_20261005